Shopping list

Quantities cover **both** drones
Each item has its own card on the following pages

PHASE 1 · buy now — Sim Lim Tower — 10 Jalan Besar · components

		ITEM	QTY	KEY SPEC
☐		**1. Addressable LED strip**	1 metre	WS2812B-2020-V6, 60 LED/m, bare, 3 wires
☐		**2. Cooling fan**	2 (one per drone)	30×30×10 mm, 5 V, 2 wires, ≤ 0.15 A
☐		**3. JST SH plug with wires (for J2)**	4	JST SH 1.0 mm, 2-pin, pre-crimped wires
☐		**4. Resistors**	10	330 Ω (or 470 Ω), 1/4 W
☐		**5. Capacitors**	4 × 470 µF + 6 × 100 nF	470 µF 10–16 V electrolytic + 100 nF ceramic
☐		**6. Silicone hook-up wire**	~2 m each, 3 colours	26 AWG silicone, stranded, 3 colours
☐		**7. Kapton tape + heat-shrink**	1 roll + assortment	Kapton ~10 mm; heat-shrink 1.5–6 mm
☐		**8. LED strip connector**	3 pairs	JST-SM 3-pin LED connector pairs

PHASE 1 · buy now — Sim Lim Square — 1 Rochor Canal Road · hobby / FPV

		ITEM	QTY	KEY SPEC
☐		**9. Propellers**	at least 2 sets of 4	Gemfan 7037-3: 7 in, 3.7 pitch, 3 blades, 5 mm hole

PHASE 2 · don't buy yet — only if phase 1 fails

		ITEM	QTY	KEY SPEC
☐		**10. 5 V step-down (buck) module**	2 (+1 spare)	MP1584 (or Mini-360), 5 V out, ≥ 1 A
☐		**11. Level-shifter chip**	2 (+1 spare)	74HCT125, DIP-14
☐		**12. Peripheral power bus**	XT30 parts · wire · JST-PH	XT30 pairs + XT30 1-to-2 splitter; 20 AWG silicone
☐		**13. Resettable fuses (PTC)**	2 each: 0.5 A, 0.65 A, 0.2 A	PTC fuses 0.5 A, 0.65 A, 0.2 A hold, ≥ 30 V
☐		**14. Fan alternatives**	1 each, only if needed	12 V or 24 V 3010 fan for the battery; 3010 blower

Bring

This sheet · the **printed prop gauge** and **30 × 30 × 10 fan block** · a ruler or calipers

Every purchase

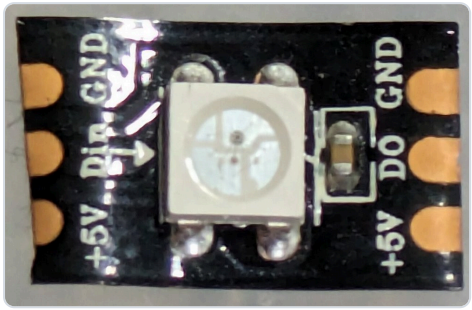
Ask each shop for a quote, and for an **itemised receipt** with the shop's name and GST number when you buy.

1 · LED strip

1 Addressable LED strip Sim Lim Tower

Buy: 1 metre

covers both drones (2 × 16 LEDs) with spares



One bare WS2812B segment: 3 pads at each cut (+5V, Din/DO, GND)

Photo: John McMaster, Wikimedia Commons, CC BY 4.0

ASK FOR

“Do you have WS2812B V6 addressable LED strip, the version that runs on 3.3 volts? The narrow kind with 2020 size LEDs, 60 per metre, no silicone sleeve.”

SPEC	MUST BE	WHY
Chip	**WS2812B V6**: 5050 (WS2812B-B-V6), 3535 (Mini-V6) or 2020	Their datasheets support a **3.3 V supply**: in spec on the drone's 3.3 V pad. Test: all LEDs off draws only µA
Wires	**3**: +5 V, data, ground	4-wire strips need a clock line the drone doesn't have
Voltage	Sold as a “5 V” strip; the chip must be **V6**	WS2812C and older WS2812B (3.7 V min) and SK6812 (3.5 V) run below spec at 3.3 V; 12 V strips won't light at all
LED spacing	**60 LEDs per metre** (16.7 mm apart)	4 LEDs fit on each ~75 mm arm
Width	**10 mm or less**; 2020 strips are ~4–5 mm	The arms are about 10 mm wide
Finish	**Bare** (IP30), no silicone sleeve or tube	A sleeve makes it ~14.5 mm wide and heavier
Colours	**RGB**	The RGBW strips use SK6812, not rated for 3.3 V

✗ **Do not buy:** WS2812C, SK6805 or SK6812 strips (not rated for 3.3 V) · APA102, DotStar, SK9822 or WS2801 (4 wires) · WS2815 (12 V) · Non-addressable RGB strip (whole strip one colour) · Silicone-sleeved or tube “waterproof” strip · 30 LEDs per metre (only 2 fit per arm)

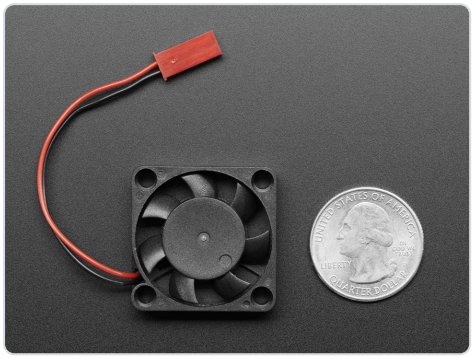
✓ **Check:** Hold the strip against a ruler: 10 LEDs should span ~167 mm, and the width must be ≤ 10 mm.

Not in any Tower shop's online catalogue (checked 2026-10-25): call Sun-Light and 123 LED Lighting first. Online fallback (check the receipt suits the Concur claim): the listing must name the V6 chip. Any WS2812B or WS2812C still works on 3.3 V, below spec.

2 · Cooling fan

2 Cooling fan Sim Lim Tower

Buy: 2 (one per drone)



Shape shown is 8 mm thick: buy 30 × 30 × 10 mm

Photo: Adafruit (product 3368)

ASK FOR

“A 30 by 30 by 10 millimetre DC fan, 5 volt, two wires only.”

SPEC	MUST BE	WHY
Size	**30 × 30 × 10 mm** (“3010”)	Fits the 36 mm-wide board stack; pushes through the 2 mm gaps in a shroud
Voltage	**5 V DC**	It runs from the VOXL 2's fan port (J2), which gives 5 V
Wires	**Exactly 2**: red (+) and black (−)	J2 switches the ground for speed control; 3- and 4-wire fans stutter
Current	**0.15 A (150 mA) or less**	J2 is limited to ~0.4 A including start-up
Bearing	**Ball** or fluid / hydraulic (“FDB”)	It will vibrate; plain sleeve bearings wear out
Plug	Any	Cut off: soldered to a JST SH plug (item 3) for J2
Noise	A **100 nF** ceramic across the fan leads (item 5)	No large capacitor: J2 switches the fan's ground
Reference part	Sunon MF30100V1-1000U-A99 (5.5 CFM, 0.12 A)	Any brand matching the rows above

✗ **Do not buy:** 12 V or 24 V fans (those are phase 2, item 14) · 3- or 4-wire fans · Sleeve-bearing fans

✓ **Check:** Compare against your printed 30 × 30 × 10 mm reference block.

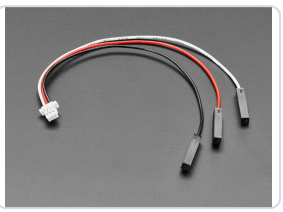
Sun-Light #03-09 online (2026-10-29): its 30×30×10 fan is 12 V, not this one. Ask around the Tower; if no shop has a 5 V one, DigiKey MF30100V1-1000U-A99-ND.

3–5 · J2 plug, resistors, capacitors

3 JST SH plug with wires (for J2) Sim Lim Tower

Buy: 4

2 drones + spares; easy to damage



Shown: a 3-pin JST SH cable. Buy 2-pin
Photo: Adafruit (product 5765)

ASK FOR
“JST SH, 1.0 millimetre pitch, 2-pin female plug, with the wires already crimped in.”

SPEC	MUST BE	WHY
Family	**JST SH**, **1.0 mm** pitch, **2-pin** (housing SHR-02V-S)	Mates with the VOXL 2's fan port J2
Wires	Pre-crimped, ~10–15 cm	Hand-crimping 1.0 mm contacts needs a special tool
If none	Any pre-made JST SH 1.0 cable to cut or de-pin (FPV shops at Sim Lim Square)	

✗ **Do not buy:** SM02B-SRSS-TB (the board-side header) · JST PH / XH / ZH / GH sizes

Sun-Light #03-09 online (2026-10-29): none. If no shop has one, DigiKey 455-3013-ND.

4 Resistors Sim Lim Tower

Buy: 10

they cost cents



330 Ω: orange-orange-brown
Photo: oomlout, Wikimedia Commons, CC BY-SA 2.0

ASK FOR
“330 ohm resistors, quarter watt.” (470 ohm is also fine.)

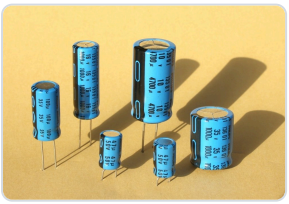
SPEC	MUST BE	WHY
Value	**330 Ω** (bands orange-orange-brown) or **470 Ω** (yellow-violet-brown)	Protects the LED data line; anything 300–500 Ω works
Rating	1/4 W, carbon or metal film, through-hole	Ordinary, smallest common size

✗ **Do not buy:** SMD chip resistors (too small to hand-solder onto wires)

Sun-Light #03-09 online (2026-10-29): 330 Ω 0.5 W, S$0.03 each.

5 Capacitors Sim Lim Tower

Buy: 4 × 470 µF + 6 × 100 nF



Various sizes shown: buy 470 µF, 10–16 V. Stripe = negative leg
Photo: Elcap, Wikimedia Commons, CC0

ASK FOR
“470 microfarad electrolytic capacitors, 10 or 16 volt, small size. And 100 nanofarad ceramic capacitors, code 104.”

SPEC	MUST BE	WHY
470 µF	10–16 V radial electrolytic, ≤ 8 mm wide	Across + / GND at the start of each LED strip; stripe = negative leg
100 nF	Ceramic, marked “104”	Across the fan leads (motor noise); spares for phase 2

✗ **Do not buy:** Large 25 V+ cans (heavy)

Sun-Light #03-09 online (2026-10-29): leaded 0.1 µF 25 V, S$0.20; no 470 µF (DigiKey P14394-ND if no shop has one).

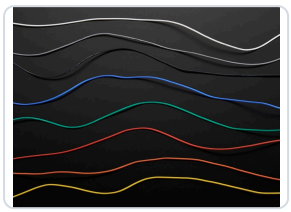
6–8 · Wire, tape, LED connector

6 Silicone hook-up wire

Sim Lim Tower

Buy: ~2 m each, 3 colours

red, black and one other colour



26 AWG silicone stranded wire
Photo: Adafruit (product 1970)

ASK FOR
“26 AWG silicone wire, stranded, in red, black and one other colour.”

SPEC	MUST BE	WHY
Gauge	**26 AWG**	Enough for the LED current; thin enough for the ESC's small solder pads
Insulation	**Silicone**	Doesn't melt back while soldering; flexes with vibration
Core	**Stranded** (many fine strands)	Solid core cracks with vibration

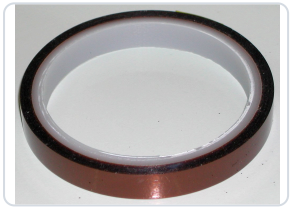
✗ **Do not buy:** PVC hook-up wire · Solid-core wire

Sun-Light #03-09 online (2026-10-29): 30 AWG silicone only, S$20.00 (acceptable).

7 Kapton tape + heat-shrink

Sim Lim Tower

Buy: 1 roll + assortment



Kapton tape: amber, thin
Photo: Jérôme, Wikimedia Commons, CC BY-SA 3.0

ASK FOR
“Kapton tape, the amber polyimide kind, about 10 millimetres wide. A small heat-shrink assortment, 1.5 to 6 millimetres.”

SPEC	MUST BE	WHY
Kapton width	**~10 mm** (6–12 mm OK), polyimide, amber	Wraps the strip ends onto the 10 mm arms; heat-proof, holds under vibration
Heat-shrink	**1.5–6 mm** assortment	Every solder joint; carbon fibre conducts
Phase 2	20 cm of **15–20 mm clear** heat-shrink	Only with the buck module: insulates it and the chip

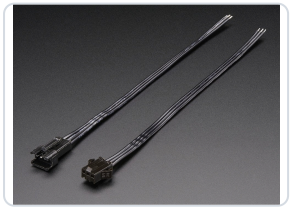
✗ **Do not buy:** Electrical PVC tape (adhesive creeps) · Heat-shrink in large sizes only (> 10 mm)

Sun-Light #03-09 online (2026-10-29): Kapton 12 mm × 30 m S$7.00; 580-piece heat-shrink kit S$16.90.

8 LED strip connector

Sim Lim Tower

Buy: 3 pairs



JST-SM 3-pin plug + receptacle pair
Photo: Adafruit (product 1663)

ASK FOR
“3-pin JST SM LED strip connector pairs, plug and socket with wires.”

SPEC	MUST BE	WHY
Type	**JST-SM 3-pin**, latching (the plug LED strips come with)	The whole LED ring unplugs from the ESC lead

✗ **Do not buy:** Dupont jumpers (shake loose)

Sun-Light #03-09 online (2026-10-29): 2-pin pairs only (S$2.50); ask around for 3-pin.

9 · Propellers

9 Propellers Sim Lim Square

Buy: at least 2 sets of 4

each set is 2 clockwise + 2 counter-clockwise; props are consumables



The bag to look for: Gemfan 7037-3
Photo: OddityRC, Gemfan 7037-3 listing



7037-3: three blades, 5 mm centre hole
Photo: OddityRC, Gemfan 7037-3 listing

ASK FOR
“Gemfan 7037, three-blade, 7 inch, 5 millimetre centre hole.”

SPEC	MUST BE	WHY
Diameter	**7 inch** (178 mm)	Same as the drone's stock props
Pitch	**3.7** (3.5 acceptable) — **never higher**	Higher pitch overheats the small motors and the ESC
Blades	**3**	Matches stock
Centre hole	**5 mm** (M5)	Fits the motor shaft
Hub	~7 mm thick	Fits the prop nut
Rotation	Set must include both **CW and CCW**	Two motors spin each way
Label	“7037-3” or “7×3.7×3”	First number = diameter, second = pitch

✗ **Do not buy:** 7040, 7042, 7045 or any “freestyle” prop (pitch too high) · Two-blade props (e.g. 7035-2) · 6 or 8 inch props · Props without a 5 mm centre hole

✓ **Check:** Lay it on your printed prop gauge: tips reach the outer ring; the hole fits the gauge's 5 mm hub.

Genuine ModalAI part: MRP-D0012-1-00 (US$29.99 / 4), online only.

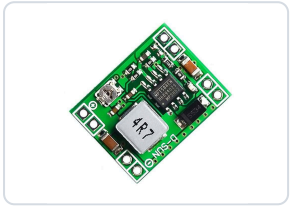
10–11 · Buck module, level shifter

Phase 2 — only if phase 1 fails. If the LED strip misbehaves on the drone's 3.3 V pad, it moves to its own 5 V buck module with a 74HCT125 on the peripheral power bus. If the fan causes trouble on J2, a 12 V or 24 V fan can run straight off the battery through the bus with a 0.2 A fuse.

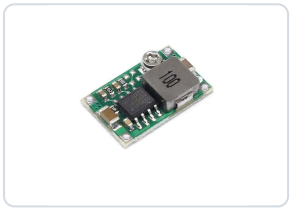
10 5 V step-down (buck) module

Sim Lim Tower · Phase 2 · only if needed · **Buy: 2 (+1 spare)**

one per drone: powers the LED strip and the fan



MP1584EN (~22 × 17 mm)
Photo: Kunkune



Mini-360 (~17 × 11 mm)
Photo: Kunkune

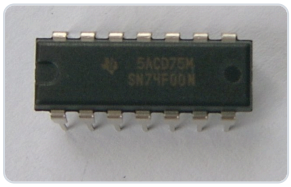
ASK FOR
“An MP1584 step-down buck converter module. A Mini-360 is also fine.”

SPEC	MUST BE	WHY
Output	**5.0 V**, **1 A or more** (MP1584 ~3 A)	Powers the LED strip and fan, separately from the flight computer
Input	Accepts **≥ 17 V** (MP1584 28 V, Mini-360 23 V max)	Runs from the 4S battery, up to 16.8 V
Size	MP1584 ~22 × 17 mm; Mini-360 ~17 × 11 mm	Light; mounts at an arm root
Adjustable?	Trim screw: **set 5.0 V with a multimeter** before use	Often shipped at another voltage

✗ **Do not buy:** Linear 7805 regulators (would burn ~6 W as heat) · Big LM2596 boards (heavy)

11 Level-shifter chip

Sim Lim Tower · Phase 2 · only if needed · **Buy: 2 (+1 spare)**



Shown: a 74-series DIP-14 chip. Buy 74HCT125
Photo: Stefan Riepl, Wikimedia Commons, public domain

ASK FOR
“74HCT125, the quad buffer chip, in DIP-14.” (The letter T matters.)

SPEC	MUST BE	WHY
Part	**74HCT125** (or 74AHCT125)	Raises the drone's 3.3 V LED data signal to 5 V for the strip
Package	DIP-14 (through-hole)	Easy to hand-solder onto a scrap of perfboard
With it	a **100 nF** ceramic capacitor (item 6) and a small perfboard scrap	Standard decoupling for logic chips; keeps the assembly tidy

✗ **Do not buy:** 74HC125 or 74LS125 (wrong input levels)

✓ **Check:** If there's no 74HCT125 anywhere: RS Singapore / element14 Singapore stock it online.

12 · Peripheral power bus

12 Peripheral power bus

Sim Lim Square | Phase 2 · only if needed | **Buy: XT30 parts · wire · JST-PH**

4 XT30 male + female pairs; 2 × 1-to-2 parallel splitters; 20 AWG silicone, 0.5 m each of red and black; 6 JST-PH 2-pin pairs



1 in → 2 out, parallel. A short cable “Y” version is also fine

Photo: Pyrodrone (XT30 Parallel Adapter)



XT30 plug with leads: buy male + female

Photo: Rotor Riot (XT30 Pigtail)

ASK FOR

“XT30 connectors, male and female pairs, and XT30 one-to-two splitter leads. Also 20 AWG silicone wire, red and black.”

SPEC	MUST BE	WHY
XT30	The small **yellow** FPV battery plug (half the size of an XT60); male + female	Rated ~15 A; the standard small drone power plug
Splitter	**1 × XT30 in → 2 × XT30 out**, parallel	Branch 1: LED/fan buck. Branch 2: future companion
Wire	**20 AWG silicone**, red + black	Battery-side bus, sized for a future companion (~1.5 A)
Plugs	**JST-PH 2.0 mm 2-pin** pre-crimped pairs × 6	Fan ↔ supply; ESC LED pads ↔ level shifter

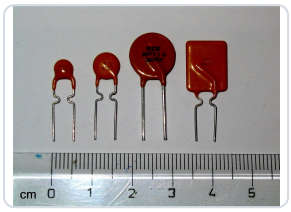
✗ **Do not buy:** XT60 (too big) · Series splitters (must be parallel)

✓ **Check:** The splitter's single side plugs into the bus pigtail from the ESC; its two sides feed the LED/fan buck and (later) the companion. Both outputs must carry the same + and −.

13–14 · Fuses, fan alternatives

13 Resettable fuses (PTC)

Sim Lim Tower | Phase 2 · only if needed | **Buy: 2 each: 0.5 A, 0.65 A, 0.2 A**



Radial PTC fuses: small dipped discs, size grows with rating (cm ruler)
Photo: Shaddack, Wikimedia Commons, public domain

ASK FOR
“Resettable PTC fuses, radial: 0.5 amp, 0.65 amp and 0.2 amp hold current, 30 volt or more.”

SPEC	MUST BE	WHY
LEDs only	**0.5 A hold** (trips ~1 A), e.g. MF-R050 / RXEF050	LED strip at the brightness cap 80/255 (~0.32 A)
LEDs + fan	**0.65 A hold** (trips ~1.3 A), e.g. MF-R065 / RXEF065	Both on one fuse (~0.44 A)
Fan only	**0.2 A hold** (trips ~0.4 A), e.g. MF-R020 / RXEF020	5 V fan (0.12 A), or a 12/24 V fan on the battery
Voltage	**30 V or more** (MF-R: 60 V)	A fan on the raw battery sees up to 16.8 V

✗ **Do not buy:** One-shot glass or blade fuses (not resettable)

Note: For tapping a shared 5 V rail, or a fan straight off the battery.

14 Fan alternatives

Sim Lim Tower | Phase 2 · only if needed | **Buy: 1 each, only if needed**



Blower: side outlet. Must be 30 × 30, 2 wires
Photo: The Pi Hut

ASK FOR
“A 30 by 30 by 10 fan, 12 volt, two wires, whose label says it runs up to about 17 volts. Or a 24 volt one. And a 30 millimetre blower fan.”

SPEC	MUST BE	WHY
12 V fan	3010, 2 wires; **label range must reach ~17 V**	Runs straight off the 4S battery (12–16.8 V) via the power bus; full charge is 40 % over a plain 12 V rating
24 V fan	3010, 2 wires; starts at ≤ 12 V	In spec across the whole 4S range, just slower
Blower	30 × 30 × 10 mm “snail shell”, 2 wires	Pushes harder into a single narrow gap
Fuse	0.2 A PTC, 30 V+ (item 13)	Any fan on the raw battery must be fused

✗ **Do not buy:** Plain 12 V fans rated to 13.8 V only (overvolted on a full pack)

Where to go

Call ahead: stock and unit numbers change

Sim Lim Tower — 10 Jalan Besar — electronic components

SHOP	UNIT	TRY HERE FOR
J.C.W. Electronic	—	Fans — **call first**; LED strip
Sun-Light Electronics	#03-09	LED strip — **call first**, 6299 8803. Online it lists only a 30 LED/m 5050 WS2812B (no V6): a bench stopgap
123 LED Lighting	#03-41	LED shop — **call first**, 8793 9683. Online catalogue is 12 / 24 / 230 V lighting strip only; ask for WS2812B V6 (5050), 5 V
Davjones Technology	#B1-47/11	Phase 2: buck modules, 74HCT125, PTC fuses
Continental Electronic Enterprise	#B1-25	Resistors, capacitors, JST connectors, wire, tape, heat-shrink

From a shop directory about two years old. No online listing confirms the fans, blower or 74HCT125 in stock anywhere in Sim Lim — call first.

Sim Lim Square — 1 Rochor Canal Road — hobby / FPV

SHOP	UNIT	TRY HERE FOR
Hobby Square	#02-60	**Propellers**; phase 2 XT30 bus parts; also silicone wire and JST SH cables if the Tower has none. Listed hours Mon–Sat 12:30–7:30 pm

If Hobby Square has no 7037 props

SHOP	WHERE	NOTES
Sky Hobbies	Jalan Sultan	skyhobbies.com.sg · large HQProp stockist
Helicopter Smartfly	Queensway Shopping Centre	smartfly.com.sg · Mon–Sat 12–7 pm, Sun 3–7 pm, closed Wednesday
Rotor Hobby	Kitchener Complex, French Road	shop.rotor.com.sg · +65 6296 3686
OddityRC	Paya Lebar Square (pickup)	oddityrc.com · stocks Gemfan 7037-3 online; allow 5–8 days

Other fallbacks

Fans	Amicus, 8 Ubi Road 2 #03-13 Zervex · +65 6980 6783 · 5 V fans by order; call first
Online, Singapore	Cytron SG and Kuriosity.sg (LED strip, buck modules) · RS Singapore and element14 Singapore (74HCT125, Sunon fans; fast delivery) · DigiKey (a few days)
Online, blower	3D-printer parts sellers and local Shopee sellers (“3010 blower 5V”)